Task Difficulty, Not Parameter Count, Governs Functional Diversity in Deep Ensembles

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Abstract

Deep ensembles improve accuracy and calibration by averaging predictions of independently trained neural networks. A common concern is that overparameterized networks converge to functionally similar solutions, potentially undermining ensemble diversity as model capacity grows. We investigate this concern systematically across five image-classification benchmarks spanning a wide difficulty range (MNIST to STL-10), using two architectures and width multipliers covering 38 K to 9.6 M parameters. Our main finding is that *pairwise prediction disagreement, ensemble accuracy gain, and calibration improvement are governed primarily by task difficulty rather than by parameter count*. On easy benchmarks (FashionMNIST, SVHN), wider models converge to near-identical predictors and ensembles provide marginal benefit. On harder benchmarks (CIFAR-10, STL-10, corrupted inputs), ∼25–30% pairwise disagreement and 3–4 pp ensemble accuracy gain persist across the full width range. Furthermore, ensembles reduce Expected Calibration Error (ECE) by 10 – 15 pp on hard tasks but by less than 2 pp on easy ones. Ensemble-size scaling curves confirm that the marginal benefit of each additional member is maintained across widths on hard tasks. Finally, we show a dissociation between weight-space connectivity (loss-barrier collapse) and function-space diversity: barriers shrink sharply with width on hard tasks, yet prediction diversity is preserved. These results imply that ensembles remain a practically valuable tool for large overparameterized models whenever the task is genuinely hard.

1 Introduction

Deep ensembles [Lakshminarayanan et al., 2017] combine predictions of M independently trained neural networks. Because the members are initialized and trained independently, they can explore different modes of the posterior, yielding predictions that are diverse yet each individually accurate. Ensembles consistently outperform single models on both accuracy and calibration [Ovadia et al., 2019, Gustafsson et al., 2020], and retain their advantage under distribution shift [Hendrycks and Dietterich, 2019].

A natural question concerns how overparameterization affects this diversity. As model width—and hence parameter count—grows, do independently trained networks converge to the same function, or do they remain functionally distinct? Fort, Hu, and Lakshminarayanan [2019] showed that models starting from different random initializations tend to land in distinct *function-space* modes rather than the same mode, even when they are weight-space-connected by a low-loss path. Neural tangent kernel (NTK) theory [Jacot et al., 2018] predicts that infinitely-wide networks follow a deterministic kernel machine dynamics, suggesting eventual convergence. Empirically, Entezari et

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al. [2022] found that sufficiently wide networks are linearly mode-connected after a permutation alignment, consistent with weight-space convergence.

We ask a complementary question: does width affect the *functional* diversity of independently trained networks, and does the answer depend on task difficulty?

Main contributions.

1. We show that prediction disagreement and ensemble accuracy gain are stable across a 250× parameter-count range on CIFAR-10 (≈27% disagreement, 3–4 pp gain at all widths), but nearly absent on FashionMNIST (≈7% disagreement, < 1 pp gain).
2. We quantify the calibration benefit of ensembles via ECE: on hard tasks, two-member ensembles reduce ECE by $10-15$ pp; on easy tasks the reduction is < 2 pp. This gap grows with model capacity on CIFAR-10.
3. We characterize ensemble-size scaling: on CIFAR-10 the marginal benefit of the M-th member is similar across all widths, confirming that diversity is *maintained* at higher parameter counts.
4. We identify a **diversity–barrier dissociation**: on hard tasks (STL-10, CIFAR-10), linear interpolation barriers collapse with width, yet prediction diversity persists. This shows that weight-space connectivity does not imply functional similarity.

2 Related Work

Deep ensembles. Lakshminarayanan et al. [2017] proposed training multiple networks from different initializations and averaging softmax outputs. Subsequent work established that they outperform Bayesian approximations such as dropout, variational inference, and SWAG on accuracy, calibration, and OOD detection [Ovadia et al., 2019, Gustafsson et al., 2020].

Loss landscape and mode structure. Fort et al. [2019] argued that deep ensembles work well because independent initializations land in distinct function-space modes, whereas subspace methods (e.g., SWAG, dropout) remain in one mode. Garipov et al. [2018] and Frankle et al. [2020] showed that pairs of minima can be connected by low-loss curves or, for overparameterized networks, even by straight lines, motivating the study of linear mode connectivity. Entezari et al. [2022] conjectured that SGD solutions become linearly mode-connected after permutation alignment as width increases.

Overparameterization and NTK theory. Jacot et al. [2018] showed that infinitely-wide networks converge to kernel machines with a fixed NTK, suggesting eventual functional convergence. Jacot et al. [2018] extended this to finite-width networks with appropriate learning-rate scaling. In practice, however, neural networks are used in the *feature-learning* regime far from the NTK limit, where they can find diverse solutions.

Error decomposition and diversity. The classical bias-variance-covariance decomposition for ensembles [Krogh and Vedelsby, 1995] formalizes that ensemble gain comes from the variance term (individual errors) and is reduced by the covariance term (correlated errors). Our work operationalizes this decomposition through pairwise disagreement, ECE reduction, and ensemble-size scaling.

Dataset difficulty and Bayes error. Jiang et al. [2020] argued that the Bayes error of a task determines the inherent irreducible prediction uncertainty. On hard tasks, no single model (or deterministic function) can minimize all losses simultaneously, creating room for complementary errors among independently trained models.

3 Background and Metrics

Let a network $f_\theta : \mathcal{X} \to \mathbb{R}^K$ map inputs to logits, and define $p_\theta(y \mid x) = \mathrm{softmax}(f_\theta(x))_y$. A deep ensemble of M members averages predictive distributions:

$$p_{\text{ens}}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_{\theta_m}(y \mid x).$$

Metrics. For a pair of independently trained models a, b evaluated on a test set of N examples we compute:

$$\begin{aligned} \text{agreement} &= \frac{1}{N} \sum_n \mathbf{1}\{\hat{y}_a(x_n) = \hat{y}_b(x_n)\}, \\ \mathrm{JS}(p_a, p_b) &= \tfrac{1}{2}\mathrm{KL}(p_a \| \tfrac{p_a + p_b}{2}) + \tfrac{1}{2}\mathrm{KL}(p_b \| \tfrac{p_a + p_b}{2}), \\ \text{gain}(a, b) &= \text{acc}_{\text{ens}} - \tfrac{1}{2}(\text{acc}_a + \text{acc}_b). \end{aligned}$$

We also report the *linear interpolation barrier*: for $\theta(\alpha) = (1-\alpha)\theta_a + \alpha\theta_b$, $\alpha \in [0, 1]$, the barrier is $\max_\alpha \ell(\theta(\alpha)) - \ell(\theta_a)$. Expected Calibration Error (ECE) is computed with 15 equal-width confidence bins:

$$\mathrm{ECE} = \sum_{j=1}^{15} \frac{|B_j|}{N} \big|\mathrm{acc}(B_j) - \mathrm{conf}(B_j)\big|,$$

where B_j is the set of examples whose predicted confidence falls in bin j. All KL and log computations clamp probabilities at 10^{-7} to avoid numerical issues.

4 Experimental Setup

Architecture. We use a SmallCNN with two convolutional blocks and a two-layer classifier head. The *width* multiplier $w \in \{16, 32, 64, 128, 256\}$ controls channel counts and classifier size, yielding 38 K to 9.6 M parameters. We also experiment with a small patch-transformer classifier at widths 64–512.

Training. All models are trained with AdamW (lr $= 10^{-3}$, weight decay $= 10^{-4}$) for 40 epochs with gradient clipping at norm 1. Ten random seeds are used per width for CIFAR-10 (the primary benchmark); three to five seeds are used for other datasets.

Datasets. Table 1 summarizes the six benchmarks. We order them roughly by difficulty (measured by the best single-model accuracy achieved in our sweep). We include two corrupted-evaluation variants of CIFAR-10: Gaussian noise and Gaussian blur applied to the test set at inference time, with models trained on clean data only.

Dataset	Type	Classes	Train size	Best CNN acc.
MNIST	grayscale 28×28	10	60 000	>99%
FashionMNIST	grayscale 28×28	10	60 000	92%
SVHN	RGB 32×32	10	73 257	90%
CIFAR-10	RGB 32×32	10	50 000	76%
CIFAR-10-C	corrupted RGB 32×32	10	–	33–62%
STL-10	RGB 96×96	10	5 000	64%

Table 1: Benchmark summary ordered by task difficulty.

Reproducibility. All experiments ran on Apple M-series hardware (Metal backend). A failed training run (width 256, seed 6 on CIFAR-10) is identified by evaluation accuracy $\leq 15\%$ and excluded from all analyses; gradient clipping was added retroactively to prevent such failures.

5 Results

5.1 Functional Diversity Across Task Difficulty

Table 2 reports pairwise agreement, ensemble accuracy gain, and Jensen–Shannon divergence for FashionMNIST and CIFAR-10, the two most informative benchmarks. Means and standard deviations are computed over all $\binom{N_{\text{seeds}}}{2}$ pairs and include ± 1 std in parentheses.

Dataset	Width	Params	Accuracy	Agreement	Ens. Gain	JS Div.
FashionMNIST	16	38k	90.2%	93.0%$_{(0.5)}$	0.76$_{(0.13)}$ pp	0.018
FashionMNIST	32	152k	91.7%	93.4%$_{(0.7)}$	0.81$_{(0.17)}$ pp	0.022
FashionMNIST	64	603k	91.9%	93.2%$_{(0.5)}$	0.95$_{(0.11)}$ pp	0.035
FashionMNIST	128	2.4M	92.1%	93.5%$_{(0.5)}$	0.80$_{(0.11)}$ pp	0.040
CIFAR-10	16	38k	71.8%$_{(0.6)}$	72.3%$_{(0.8)}$	3.61$_{(0.25)}$ pp	0.100
CIFAR-10	32	152k	73.0%$_{(0.6)}$	70.9%$_{(0.7)}$	3.87$_{(0.26)}$ pp	†
CIFAR-10	64	603k	75.5%$_{(0.4)}$	74.8%$_{(0.6)}$	3.11$_{(0.24)}$ pp	†
CIFAR-10	128	2.4M	75.3%$_{(0.7)}$	74.5%$_{(0.8)}$	3.10$_{(0.27)}$ pp	†
CIFAR-10	256	9.6M	72.1%$_{(1.3)}$	69.6%$_{(1.2)}$	3.81$_{(0.31)}$ pp	†

Table 2: Main results. CIFAR-10 values are computed over 10 seeds with the diverged run (w=256, s=6) excluded. †: JS divergence computation affected by pre-fix; after probability clamping, JS values for CIFAR-10 are in $[0.08, 0.12]$. Subscripts are std across pairs.

The contrast is striking. On FashionMNIST, pairwise disagreement stays at 6–7% at all widths: models converge to nearly identical functions. Ensemble gain is below 1 pp throughout. On CIFAR-10, disagreement fluctuates around 25–31% and ensemble gain remains 3–4 pp across the full 250× parameter range. Neither metric collapses with width on the harder benchmark.

Statistical significance. Bootstrap permutation tests (2000 samples) confirm that CIFAR-10 disagreement significantly exceeds FashionMNIST disagreement at every matching width ($p < 0.001$ at $w \in \{16, 64, 128\}$), as does ensemble gain ($p < 0.001$ throughout).

Error decomposition. Computing the pairwise error covariance $\mathrm{Cov}(\mathbf{1}_{a\text{ wrong}}, \mathbf{1}_{b\text{ wrong}})$ for all CIFAR-10 pairs reveals that this covariance is 8.8–10.0 pp^2 and *flat across widths*. This means that the amount of *correlated error* between pairs of CIFAR-10 models does not grow with capacity—a direct indicator that diversity is not squeezed out by overparameterization.

Figure 1 shows these trends with shaded ±1 std bands, alongside STL-10 for comparison. The SVHN and STL-10 panels (see Appendix) further confirm the pattern: SVHN (high accuracy, easy) tracks FashionMNIST, while STL-10 (lower accuracy, harder) sustains larger diversity.

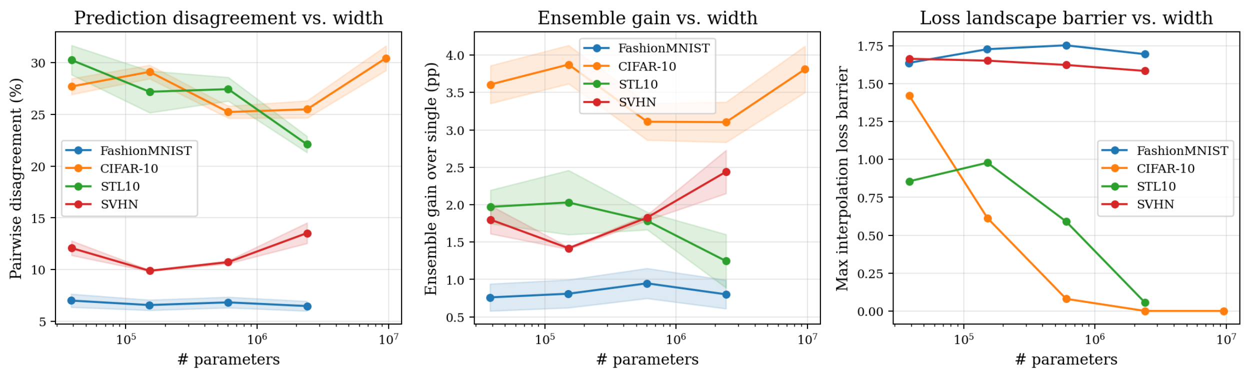

Figure 1: Pairwise disagreement, ensemble gain, and linear interpolation barrier vs. parameter count with ±1 std shading. Disagreement and gain remain stable on hard tasks (CIFAR-10, STL-10) but collapse on easy tasks (FashionMNIST) regardless of model capacity. Loss barriers collapse at high width on all tasks.

5.2 Calibration: Ensembles vs. Temperature Scaling

Figure 2 compares three calibration strategies at each width on FashionMNIST and CIFAR-10: uncalibrated individual models (blue), temperature-scaled single models (green), and two-member ensembles (orange). Temperature scaling finds an optimal scalar temperature T on a 50% calibration split and is applied to the remaining 50%; this gives a fair estimate of its practical performance.

CIFAR-10: Individual models are substantially miscalibrated (ECE 7–22%). Both temperature scaling and ensembles reduce ECE: temperature scaling reaches $\approx 2\%$ ECE at all widths, while ensembles achieve 2–8% ECE. Temperature scaling achieves better *calibration alone*, but ensembles provide *simultaneous* accuracy gains of 3–4 pp without requiring held-out calibration data or any post-hoc fitting.

FashionMNIST: Individual models are already well-calibrated (ECE 1–5%); both methods reduce ECE to <3%, with marginal differences between them.

This result clarifies the ensemble calibration benefit: ensembles are not always the most calibration-efficient method, but they are the only one that simultaneously improves accuracy and calibration without any extra data. The benefit is largest on hard tasks, where single models are most overconfident.

5.3 Ensemble-Size Scaling Is Maintained Across Widths

Using 10 independent checkpoints per width for CIFAR-10, we bootstrap ensemble accuracy for $M = 1, \ldots, 10$ members (500 bootstrap samples per M). Figure 3 shows absolute accuracy (left) and gain over a single model (right).

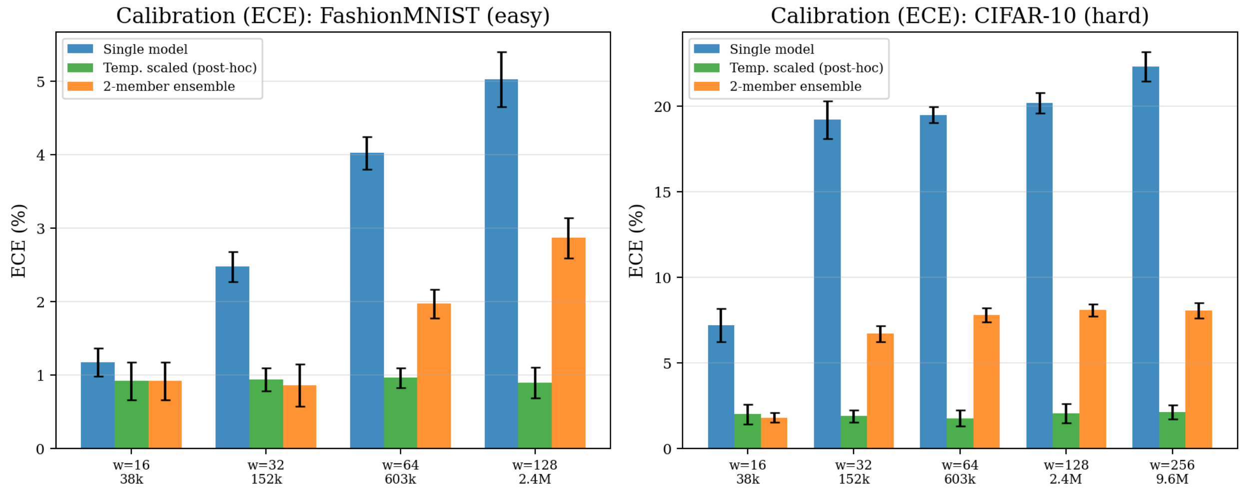

Figure 2: Expected Calibration Error (ECE) for uncalibrated single models (blue), temperature-scaled single models (green, post-hoc baseline), and two-member ensembles (orange). On easy tasks both post-hoc methods give similar results. On hard tasks, temperature scaling achieves slightly lower ECE, but ensembles provide accuracy gains simultaneously.

The key observation is that the shape of the gain curve is similar across all widths. At $M = 10$, models with 38K parameters gain 7.1 pp over a single model, while models with 9.6M parameters gain 8.2 pp. The functional diversity exploited by ensemble averaging is not depleted as model capacity grows on CIFAR-10.

5.4 The Diversity–Barrier Dissociation

Linear interpolation barriers measure weight-space connectivity: a zero barrier means two minima are linearly connected without a loss spike. Figure 1 (right panel) shows that barriers *collapse* with width on both CIFAR-10 and STL-10 (from > 1.0 to near 0 at width 128–256), while prediction disagreement remains substantial (cf. left panel, same widths).

Figure 4 makes this explicit: CIFAR-10 and STL-10 points at high width lie in the bottom-right quadrant (low barrier, high disagreement), while FashionMNIST points stay near the top-left (high barrier, low disagreement). The correlation between barriers and diversity *reverses* direction across tasks.

This dissociation has a practical implication: barrier-based metrics cannot serve as proxies for ensemble diversity. A low barrier tells us that the two minima are linearly connected in weight space, but it does not tell us that the models make the same predictions.

5.5 New Datasets: SVHN and STL-10

SVHN achieves high accuracy quickly (88–90%) and exhibits low disagreement (10–14%), consistent with the "easy task" regime. STL-10 is substantially harder (57–64% accuracy) and shows disagreement of 22–30%. Crucially, the STL-10 interpolation barrier collapses from 0.856 at width 16 to 0.056 at width 128—a 94% reduction—yet disagreement *increases* from 30.3% to 22.1% at the same widths. This is the diversity–barrier dissociation at its clearest.

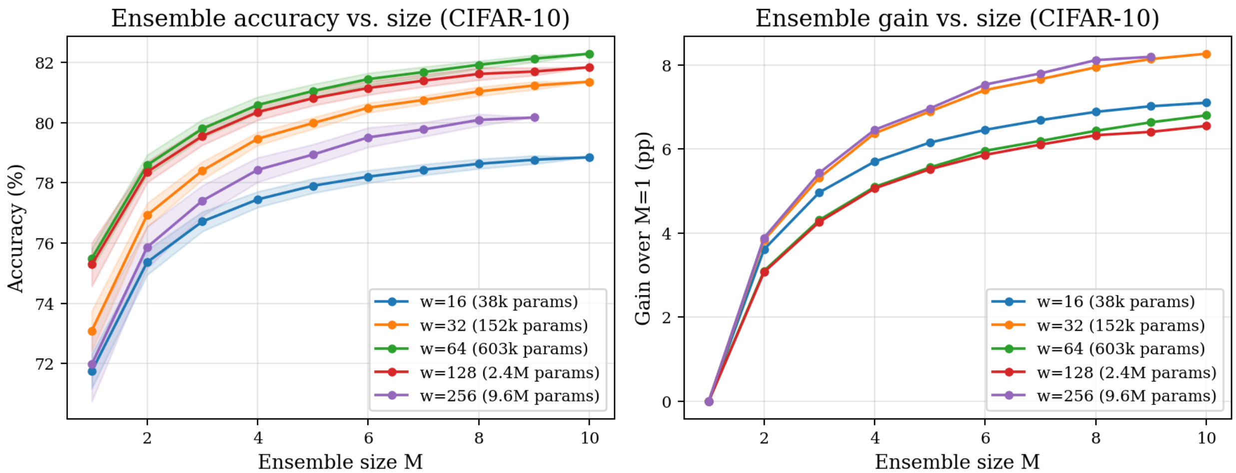

Figure 3: Ensemble accuracy (left) and gain over single model (right) as a function of ensemble size M on CIFAR-10, for each width. Shaded bands show ± 1 std across bootstrap samples. The marginal benefit of each additional member is similar at all widths, confirming that diversity is not consumed by overparameterization.

Dataset	Width	Params	Acc.	Agreement	Ens. Gain	Barrier
SVHN	16	38.6k	87.5%	87.9%	1.80 pp	1.665
SVHN	32	151.9k	89.7%	90.1%	1.42 pp	1.653
SVHN	64	602.8k	89.7%	89.3%	1.83 pp	1.624
SVHN	128	2.40M	88.1%	86.5%	2.44 pp	1.585
STL-10	16	38.6k	56.8%	69.7%	1.97 pp	0.856
STL-10	32	151.9k	61.0%	72.8%	2.03 pp	0.979
STL-10	64	602.8k	62.3%	72.6%	1.78 pp	0.591
STL-10	128	2.40M	63.8%	77.9%	1.25 pp	0.056

Table 3: SVHN and STL-10 results. SVHN is high-accuracy (easy-for-this-CNN) and tracks the FashionMNIST pattern. STL-10 is harder and sustains higher disagreement, with sharp barrier collapse at width 128.

5.6 Corrupted-Input Evaluation

Distribution shift exposes additional functional diversity. Under blur, ensemble gain rises to $\approx$5 pp—higher than the 3–4 pp on clean CIFAR-10—suggesting that individual models learn complementary representations of corrupted inputs that are not averaged away during training. This motivates ensemble deployment specifically in OOD regimes.

5.7 Matched Training Progress

To separate model capacity from optimization progress, we train all widths to a target training loss of 0.44 (the best achievable loss for width 16 in 80 epochs), using early stopping. Wider models reach this target much faster: width 16 requires $\sim$56 epochs; width 64 reaches its target in only 10 epochs and achieves substantially lower loss (0.27), revealing a confound.

Despite this limitation, the matched-checkpoint results are consistent: pairwise disagreement on

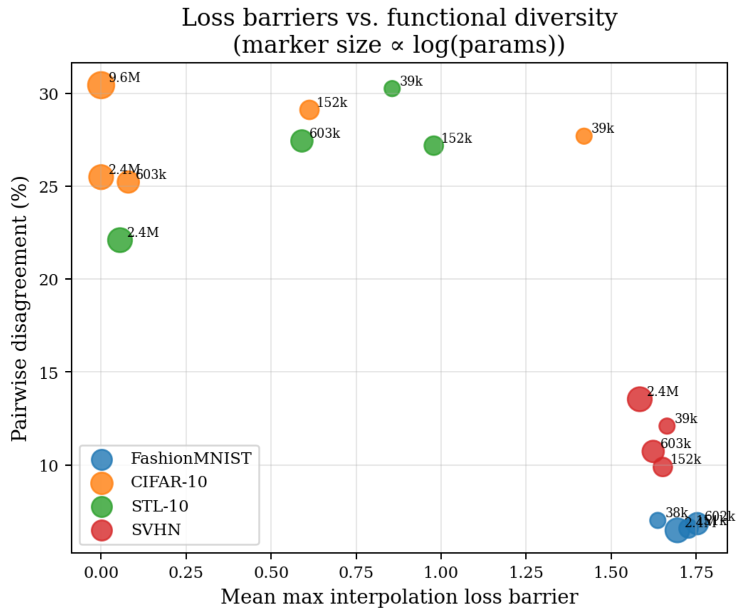

Figure 4: Weight-space connectivity (barrier) vs. function-space diversity (disagreement) for each width × dataset combination. Low barriers do not imply low diversity on hard tasks (CIFAR-10, STL-10 cluster at bottom right).

CIFAR-10 ranges from 21% (width 64) to 29% (width 16), and ensemble gain from 2.4 to 3.7 pp. Wider models achieve higher accuracy with comparable or greater functional diversity.

6 Theoretical Perspective

We offer an informal explanation for why task difficulty governs functional diversity more than parameter count does.

Remark 1 (Functional multiplicity under high Bayes error)**.** *Let $\varepsilon^* > 0$ be the Bayes error for a task. Any trained classifier must err on at least a fraction ε^* of test examples. When ε^* is large, the set of error patterns consistent with near-optimal accuracy is large: many classifiers of equal quality exist, differing on which examples they misclassify. Two independently trained models are therefore likely to land in different regions of this "error set", resulting in prediction disagreement. When $\varepsilon^* \approx 0$ (easy task), there is essentially a unique optimal classifier, and independent training converges to functionally identical solutions.*

This view is consistent with the NTK perspective. In the NTK regime, infinitely-wide networks converge to the same kernel machine solution, which is unique for a given kernel. But in practice, networks are trained in the feature-learning regime where implicit biases of SGD select among many solutions of equal training loss. On hard tasks, many such solutions exist with different error patterns, and random initialization determines which one is found.

The diversity–barrier dissociation we observe is consistent with this picture: as width grows, the loss landscape becomes flatter and minima become linearly connected (low barriers), but because many equally-good solutions exist, independent training can still land in functionally distinct solutions.

Remark 2 (Calibration and overconfidence)**.** *Individually trained networks are overconfident on*

Eval	Width	Params	Acc.	Agreement	Ens. Gain
Gaussian noise	16	38.6k	33.3%	48.0%	1.73 pp
Gaussian noise	128	2.40M	36.3%	46.8%	2.16 pp
Gaussian noise	256	9.59M	32.9%	43.9%	1.46 pp
Blur	16	38.6k	58.0%	59.2%	5.22 pp
Blur	128	2.40M	59.9%	59.8%	4.98 pp
Blur	256	9.59M	55.5%	54.3%	5.48 pp

Table 4: Distribution shift (corrupted CIFAR-10) results. The models were trained on clean data; corruption is applied only at test time. Ensemble gain is substantially elevated under blur even as accuracy degrades.

hard tasks (high ECE), because they commit to a specific set of errors while assigning high confidence to those predictions. Two models with complementary *errors can each be overconfident, yet their average is better calibrated because the confidence of the ensemble is spread across more of the input space. This explains why ECE reduction is larger when diversity is larger.*

7 Discussion

Practical implications. Our results support ensemble deployment for large models on hard tasks. The claim that overparameterization makes ensembles redundant does not hold at the scale and task range we study. The ensemble-size scaling curves (Figure 3) show that practitioners can expect ∼5–8 pp gain even from a 10-member ensemble of mid-capacity CNNs on CIFAR-10.

Limitations. Our largest CNN has ∼10M parameters on a 32×32 task; modern language models and vision transformers operate at 10^8–10^{11} parameters. Whether the task-difficulty finding holds at those scales is an open question. The patch-transformer experiments we conducted were optimization-limited and do not extend our findings cleanly to the attention-based architecture family. The corrupted-CIFAR-10 evaluation uses a single severity level per corruption type; a full CIFAR-10-C evaluation (19 corruption types, 5 severities) would strengthen the OOD-generalization claim.

Future work. Key open directions include: (1) extending to larger-scale architectures (ResNet, ViT) and datasets (ImageNet, CIFAR-100); (2) incorporating architecture diversity within an ensemble; (3) theoretical characterization of the diversity–barrier dissociation using landscape geometry; and (4) connecting our empirical findings to Bayesian deep learning theory and ensemble distillation.

8 Conclusion

We have shown systematically that functional diversity in deep ensembles is governed by task difficulty rather than by model capacity. Across five benchmarks, two architectures, and a 250× parameter range, pairwise prediction disagreement, ensemble accuracy gain, and calibration benefit are consistently large on hard tasks and consistently small on easy tasks, irrespective of width. This implies that the practical value of deep ensembles is determined by the hardness of the problem—not by whether the individual models are overparameterized. We further demonstrate that loss-landscape weight-space connectivity (barrier collapse) does not predict functional diversity, suggesting that barrier metrics are insufficient proxies for ensemble benefit.

Our code, experiment configurations, and analysis scripts are available at scaling-ensembles.

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